

FB_Inference - SCL excerpt (B-spline LUT)

```
FUNCTION_BLOCK FB_Inference
VAR_INPUT features:ARRAY[0..27]OF REAL;END_VAR
VAR_OUTPUT class_id:INT;confidence:REAL;END_VAR
FOR i:=0 TO 27 DO
    lo:=0;
    FOR j:=1 TO 13 DO
        IF features[i]>=W_DB.g0[j] THEN lo:=j;END_IF;
    END_FOR;
    t_val:=(features[i]-W_DB.g0[lo])/(W_DB.g0[lo+1]-W_DB.g0[lo]+1E-10);
    FOR o:=0 TO 15 DO
        v3[o*28+i]:=W_DB.t1[base+lo]*(1-t_val)+W_DB.t1[base+lo+1]*t_val;
    END_FOR;
END_FOR;
END_FUNCTION_BLOCK
```

Syntax-highlighted SCL rendered as print-ready vector figure.